

Activity- 2

AIM:

To study the variation in potential drop with the length of a wire for a steady current.

APPARATUS

A fully charged 4.5V battery or battery eliminator, a potentiometer, a low-resistance rheostat, a voltmeter of range (0-3V), an ammeter(0-3A), a one-way key, a jockey, a set square, connecting wires, and a piece of sandpaper.

THEORY:

For a potentiometer with a wire of uniform material density and thickness (cross-sectional area) carrying a steady current, the potential drop is proportional to the length of the wire

$$V \propto l$$

$$V = Kl$$

$$K = \frac{V}{l} = \text{constant}$$

Where K is the drop of potential per unit length, it is called the potential gradient.

PROCEDURE:

1. Draw a circuit diagram showing the scheme of connections as in the figure.
2. Remove the insulation from the ends of the connecting copper wires with sandpaper.
3. Connect the positive pole of the battery (eliminator) (a battery of constant emf) to the zero end (P) of the potentiometer and the negative pole through a one-way key, an ammeter, and a low-resistance rheostat to the other end (Q) of the potentiometer.
4. Connect the positive terminal of the voltmeter to the end P of the potentiometer and the negative terminal of the jockey.
5. Touch the end of the jockey to the end Q of the potentiometer.
6. Close the key and set the rheostat such that the voltmeter gives full-scale deflection (3V)
7. Touch the jockey at the end P at 0(zero)cm. The voltmeter will give zero deflection.
8. Touch the jockey at marks separated by a 50 cm length of wire. Note the voltmeter reading in each case.
9. Record your observations in tabular form as given.

RESULT:

For a steady current, the potential drop is directly proportional to the length of the wire.

The potential drop per unit length of wire=

Precautions

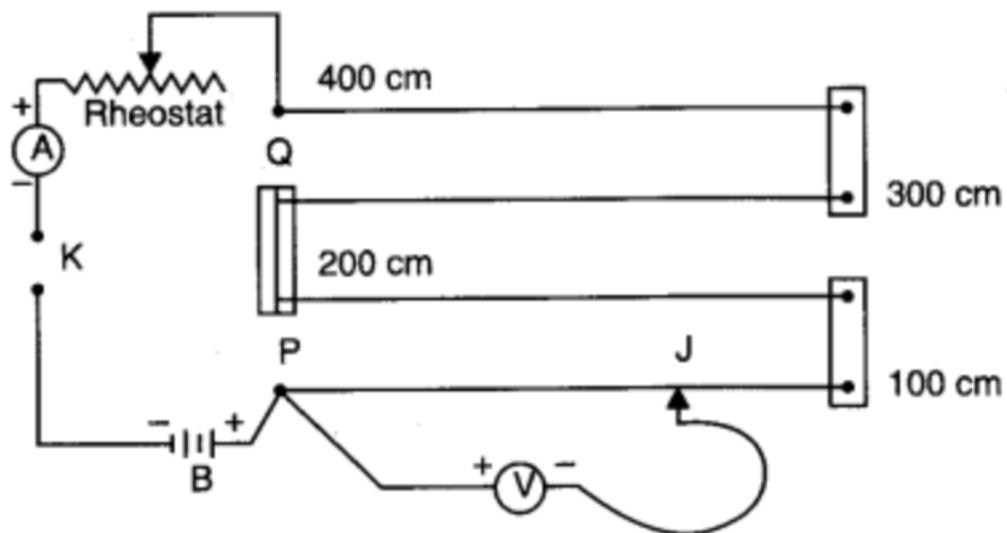
- i. The connections should be neat, clean, and tight.
- ii. All precautions regarding the polarity should be observed
- iii. Zero correction, if any, must be applied.

Sources of Error

- i. The instrument screws may be loose.
- ii. Thick connecting wires may not be available.
- iii. A rheostat may have high resistance.

Diagram and observation table on the left side of the page

Diagram:



Observations and calculations

Least count of voltmeter=

S.No.	Length of potentiometer wire (l)	Voltmeter reading (V)	Ratio, $K = \frac{V}{l}$
1			
2			
3			
4			
5			